

EC441 Week 1 problems

1. A digital signal is transmitted over a noisy channel with a bandwidth of 2 MHz. The SNR is 30 dB. Determine Shannon-Hartley Capacity

First: Convert SNR to linear

$$30 = 10 \log_{10} (\text{SNR}_{\text{lin}})$$

$$\text{SNR}_{\text{lin}} = 10^3 = 1000.$$

$$C = B \log_2 (1 + \text{SNR})$$

$$= 2 \log_2 (1 + 1000) = 19.93 \text{ Mbps}$$

2. Explain why Fiber Optic cables support higher data rates over long distance than twisted pair copper cables.

This is because light signals in high-quality fiber experience much less attenuation per kilometer, so it can travel much further, compared to electrical signals in copper. Also copper cables are susceptible to electro magnetic interference, which makes them have a lower SNR compared to fiber-optic. Fiber-optics higher SNR gives them a higher capacity.